Analytical Chemistry Cumulative Exam
April 25th, 2025
Titrations

Instructions

Please show your working and clearly indicate your final answer (e.g., underline, box). No credit will be given for answers given without working. **Partial credit will be given for each step of the solution**. A correctly set up problem containing calculator errors could be sufficient to demonstrate you understanding the material and help you pass this cume!

You should use the <u>appropriate number of significant figures</u> in all your answers.

1) 25.0 mL of a 0.0640 M EDTA solution was added to a 41.0 mL solution which contained an unknown concentration of V^{3+}. The system was allowed to react, and after doing so, all of the V^{3+} originally present in the solution was complexed by EDTA. An excess of EDTA remained in solution. This solution was back-titrated with a 0.0480 M Ga^{3+} solution until all of the EDTA reacted. This required 17.0 mL of the 0.0480 M Ga^{3+} solution. Determine the original concentration of the V^{3+} solution?

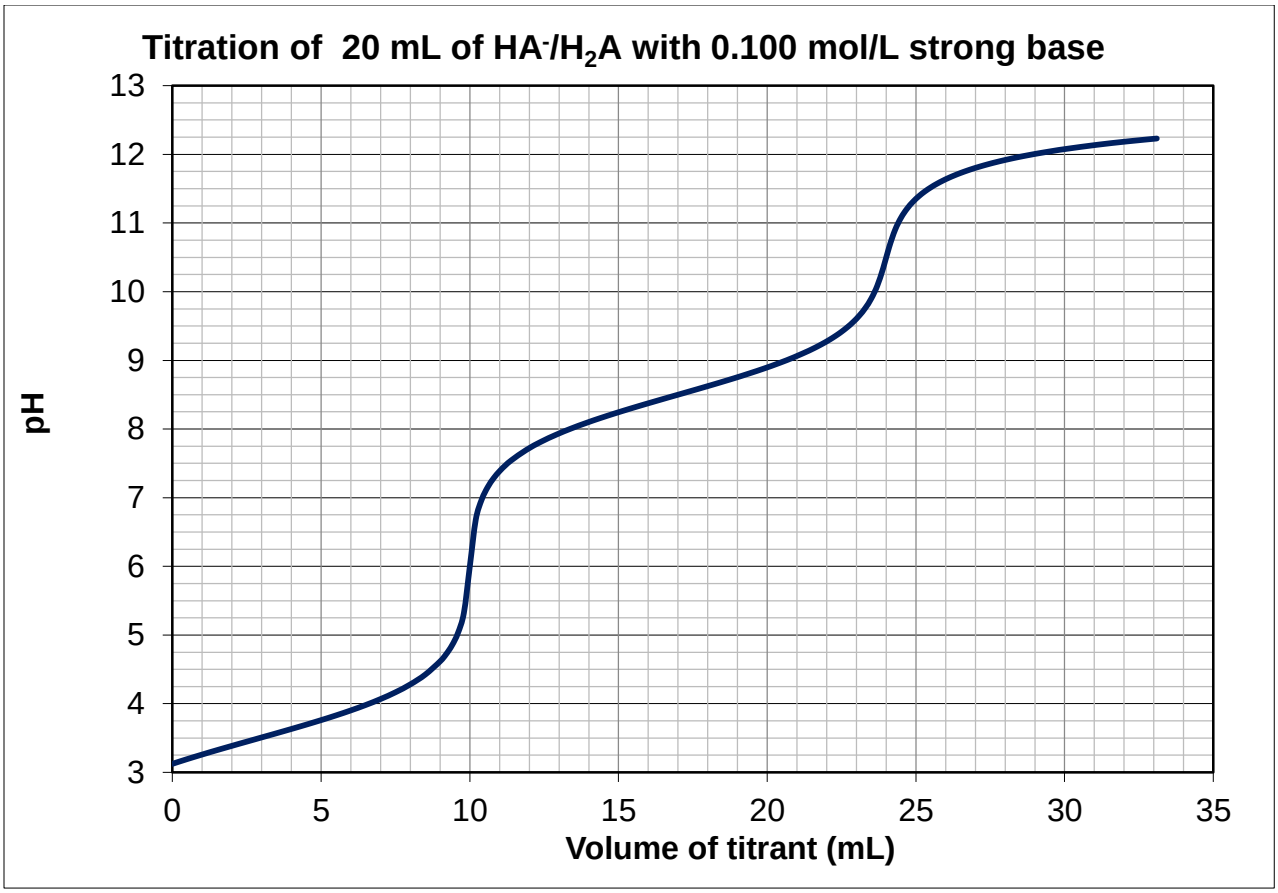
2) In a precipitation titration, a 25.00 mL solution of 0.03160 M Na_2CO_3 is titrated with 0.02790 M $Ba(NO_3)_2$. Calculate pBa^{2+} following the addition of the three volumes of $Ba(NO_3)_2$ listed. The K_{sp} for $BaCO_3$ is 5.0×10^{-9}

   a) v_e (volume at the equivalence point – you will need to determine this)
   b) 10.40 mL
   c) 37.50 mL

3) Although 'endpoint' and 'equivalence point' for a titration are sometimes used interchangeably, they have different meanings. It is important that you understand the difference. In 100 words or fewer, define each term, making it clear that you know the difference. Note: it is possible to get full points with <50 carefully chosen words.

4) The following plot (next page) shows the titration with a 0.100 M strong base of a 20 mL solution containing weak acid in two forms (HA^- and H_2A) of unknown concentrations. From the graph estimate:

   i) The initial concentrations of both the HA^- and H_2A
   ii) The pK_a values of the weak acid


Titration of 20 mL of HA-/H2A with 0.100 mol/L strong base
pH
13
12
11
10
9
8
7
6
5
4
3
0
5
10
15
20
25
30
35
Volume of titrant (mL)